



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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**Let's Start!**

I don't have a clue about programming logic

**Learning Outcome:**

Exchange information about logic, its principles, and methods in simple spoken and written situations.

Ask your partner:

- Do you think formal logic is involved in programing? Why?
- What other areas do you think are part of programming?

Can Do Statements:

- I can give a simple definition for programming logic.
- I can talk about the concept of programming logic.
- I can list some components of computational logic.





Let's Explore!

Kim helps Lilly study for the test about programming logic.



Instructions: Read the text below.

Lilly: Hi Kim, can you help me with this?

Kim: Sure, What can I do for you?

Lilly: Well, I know the final exam is coming, and I'm very confused by the entire section of programming logic. Can you help me study for this?

Kim: Okay. Let's go to the library!

Lilly: Thank you! That is so nice of you.

Kim: Tell me. What do you know about programming logic?

Lilly: I don't understand a thing! Can we start from the very beginning?

Kim: All right, let's start from the origin. Programming logic is the base of programming itself. It is a group of coherent ideas and reasons. This science studies the structure, the fundamentals, and use of the human knowledge applied to computer programming.

Lilly: Oh, I see, now I remember that the teacher said something about logical operations that work according to logical principles and quantifiable results. I guess these ones have to do with the advancement of computer science. She also says that hard data or concrete evidence construct all.

Kim: Well, it seems you know more than you think!

Lilly: I get confused with the entire logic thing.

Kim: I know. Computational logic is not easy because it involves many disciplines, I mean, there are logic processes that not only involve the computer science, therefore, this subject needs the help of more areas to be developed. For instance, we always need to rely on mathematical logic when we are programming.

Lilly: Oh, now I see. That is why the teacher also mentions the artificial intelligence and deduction systems.

Kim: That's right! Those areas are in computational logic. This subject studies the implementation of programs to discover or check formal statements. It also explains relations based on order logic. The initial programming language is Prolog.

Lilly: Thank you very much!

Kim: You're welcome.

**Let's Practice!**

Computational logic is very hard to understand!

Instructions: Select the correct option to correctly complete each sentence.

1. Let's start from the origin, (programming logic / Deduction systems / inference techniques) is the base of programming itself.
2. We always need to rely on (mathematical logic / Inference techniques / deduction systems) when we are programing.
3. The teacher would also talk about artificial intelligence and (deduction systems / knowledge representation / mathematical logic).

**Let's Practice!**

Checking the True Behind Programming Logic Concept

Instructions: Read the following sentences based on the text. In front of each sentence write True or False.

1. Programming logic studies the structure, fundamentals and use of the human knowledge applied to computer programming.
2. Logical operations work according to quantifiable results.
3. This subject involves just one discipline more.



Extension Activity

Searching deeply programming logic features.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about programming logic.
- Make notes about it and create a poster to explain the main features.
- Use simple present tense and demonstratives in your sentences.
- Share with your partners and summarize the importance of it.

Self-Assessment

Yes

No

- I search information about programming logic.
- I explain deeply its main features.
- I create graphic tools to show information

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☐	☐
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Target Words

- **programming logic** (noun): the way all human logic knowledge can be applied to computer programming
e.g. Programming logic is the base of programming itself. It is a group of coherent ideas and reasons.
- **coherent** (adjective): something that is logical and constant
e.g. Programming logic is the base of programming itself. It is a group of coherent ideas and reasons.
- **structure** (noun): a system composed of parts that are put together for a purpose
e.g. This science studies the structure, the fundamentals, and use of the human knowledge
- **quantifiable** (adjective): something that can be measured exactly
e.g. Logical operations work according to logical principles and quantifiable results.
- **hard data** (noun): Information that can be demonstrated, it can be numbers or facts
e.g. She also says that hard data or concrete evidence construct all.
- **Artificial Intelligence (AI)** (noun): the capacity of a computer to make decisions and learn similar to humans
e.g. That is why the teacher also mentions the artificial intelligence and deduction systems.



Learning Points

Demonstrative Pronouns

You use a **demonstrative pronoun** to point out something specific in a context.

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The **demonstrative pronouns** can also indicate an object in space or time, and they can be singular or plural.

When an object is near the speaker, we can use **"this"** as a singular noun and **"these"** for plural nouns.

Example.

- I want **this** t-shirt.
- I love **these** gloves in my hands.

When there is a distance between the speaker and the object, we can use **"that"** for a singular noun or **"those"** for plural nouns.

Example:

- I like to buy **that** computer over there!
- She needs to buy **those** books about programming.

There are simple rules to use demonstratives correctly:

1. They always identify situations or nouns.

Example:

- I can't believe **this!** I have a new computer!

2. We use the demonstratives to describe animals, places or things.

Example:

- **These** computers are very old! We need to change them.

Examples

- That is why the teacher also mentions the artificial intelligence and deduction systems.
- Thank you, that's so nice of you.





Learning Points

Components of Computational Logic

Computational logic refers to the interdisciplinary practice that constructs the fundamentals on computer science.

Computational logic is very wide. As most of the sciences, it has theoretical and practical elements that are based on mathematics, artificial intelligence, logic and certainly a computational science.

Some of the areas that are included in **computational logic** are:

- Programming Logic
- Deduction systems
- Knowledge representation
- Inference techniques
- Mathematical logic

Computational logic is actually the logic of the object that is common in all areas involved. There is a use of logic in all areas that have an application on **computational logic**.

Examples

- *Programming logic is the base of programming itself.*
- *Computational logic can be confusing because it involves many disciplines.*